Class. ENTRE 445/545 Health Innovation Practicum, Autumn Quarter

Class Hours. Thursdays 4-5:50 pm

Office Hours. e-mail wjc8@uw.edu to arrange.

Text: There is no required textbook. Texts will be provided from those available through the UW library. Whenever possible these will be posted for students to download from the Canvas website

Instructor.

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Course Description.

This course is designed to give students an overview of the challenges that are necessary for any young business in healthcare or life sciences to consider. By the end of the course students should expect to have an awareness of the system of regulation of health technologies, the process of development for health technologies, and the economics of healthcare.

Course Learning Objective

Healthcare is an industry that, at its best, is built around strong reproducible science with healthcare providers acting as the patient's agent. This places the consumer in a different position than other markets where they are more in control of their purchases. As such, those working in the area will need to have a grasp of the concepts of entrepreneurship as well as a strong understanding of when the traditional rules of economics and entrepreneurship no longer apply.

Teaching and Class Format

- *Before Class*- Each week students will be given a brief set of readings to review before class. These are intended to prepare students for the topic that will be covered in class and ensure that we have a productive use of time together. 2-3 videos will also be posted that cover the main points from the lecture.

- *During Class*- As much as possible class time will be designed to be interactive to make the most of our limited time together. This may include discussions of a business case, an activity performed in groups or guest lectures.

- *After Class*- After class you will be expected to complete a brief quiz to evaluate your understanding of the course material. Additionally, you may be asked to identify and interview an expert with domain expertise on the topic covered in class and an awareness of your technology area.

Class Project and Work Outside of Class

In addition to attending and participating in all classes you will also be expected to complete some work outside of class both individually and as part of a team that will be assigned to you.

- *Preparation for Class*- As mentioned above, you will be expected to prepare for class each week by completing the readings assigned for the week. This will not be graded but will give you the appropriate background to participate in class discussions and activities.

- *Domain Expert Interviews*- One of the most important activities for any entrepreneur is identifying domain experts, learning from them, and assimilating their feedback into your overall strategy. To hone this ‘soft skill’ your team will be asked to complete at least 3 interviews (~30min) with domain experts over the course of the term. You should decide as soon as possible who will be responsible for each interview so that the effort of scheduling can be spread across the team. Your team will be responsible for providing a brief report of each interview as well as incorporating the feedback from these interviews into your final report.

- *Class Project and Presentation* - Within your team you will be asked to complete a project that touches on all of the domains (background science, clinical research, regulatory considerations, economics and reimbursability, ethics) that we have discussed in class. There are 2 ways to complete this project. You can review a technology or healthcare area (pre-approved examples below, alternatives only by permission of the instructor) or you can review a single technology or business idea. You should incorporate your interviews in order to summarize your work throughout the term.

Grading

ENTRE 445/545 is offered pass/fail only. As described in the sections above, this course requires a lot of class participation, teamwork, and practical application of lecture material. DO NOT TAKE THIS CLASS IF YOU ARE NOT GOING TO PARTICIPATE AND WORK WITH OTHERS. Passing performance is based on the four elements below:

- Peer Reviews 10%
- Interview Reports 20%
- Final Presentation 30%
- Weekly quizzes 40%

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CLASS ONE OVERVIEW

Objective: Introduce the course and expectations, give students an overview of each of the technology areas, and encourage them to consider which area they will focus on for their project.

In-Class Lecture: (1 hour) A brief overview of the challenges facing the healthcare system, an overview of the course, and my expectations. Introduction of technology areas that teams can choose from for their class project.

In-Class Activity: (50 min) For the last 50 minutes of class, students will introduce themselves briefly and create project team groups. Within their group, they will be expected to introduce themselves and rank which technology they would like to work on. Topics pre-approved for this include:

Technology Areas	**Healthcare Areas**
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Synthetic Biology	Aging
Microbiome based diagnostics or treatments	Mental health
mHealth	Pricing transparency in healthcare
Internet of Things, connected bio-sensors, and wearables	Access and Affordability of healthcare services
Vaccines for cancer or infectious disease, immunotherapy	Rural health
GenAI	Women's health
Blockchain and distributed ledger technologies	Maternal health and safety
Virtual Reality and Augmented Reality	Addiction prevention and treatment
Tele-Health	Pandemic preparedness
Point-of-care diagnostics	Environmental health and safety

Technologies outside of those included on the pre-approved list must be approved by the instructor. Additionally, students may suggest a concept or business idea that they would like to work on through all of the modules for the course.

Homework:

Quiz 1- (Due October 2, 11:59pm)